

New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

Total number of customers who have placed orders:

```
SELECT
    COUNT (DISTINCT customer_id) AS total_customer_count
FROM order_t;
```

Distribution of the customers across states:

```
SELECT
    state, COUNT (DISTINCT customer_id) AS customer_count
FROM customer_t
GROUP BY state
ORDER BY customer_count DESC;
```

Distribution of the customers who have placed orders across states:

```
SELECT
    c.state, COUNT (DISTINCT o.customer_id) AS customer_count
FROM customer_t AS c
INNER JOIN order_t AS o
    ON c.customer_id = o.customer_id
GROUP BY c.state
ORDER BY customer_count DESC;
```

Output:

Total number of customers who have placed orders:

SQL queries passed

Query 1 (Passed): SELECT COUNT(DISTINCT customer_id) AS total_customer_count FROM order_t

Output:

Showing 1 rows

total_customer_count
994

Distribution of the customers across states:

SQL queries passed

Query 1 (Passed): SELECT state, COUNT (DISTINCT customer_id) AS customer_count FROM customer_t GROUP BY state ORDER BY customer_count DESC

Output:

Showing first 10 rows out of 37 rows

state	customer_count
California	17
Texas	10
Florida	9
New York	7
Virginia	5
Michigan	5
Illinois	5
District of Columbia	5
Pennsylvania	4
Ohio	4

Distribution of the customers who have placed orders across states:

SQL queries passed

Query 1 **Passed**: SELECT c.state, COUNT (DISTINCT o.customer_id) AS customer_count FROM customer_t AS c INNER JOIN order_t AS o ON c.customer_id = o.customer_id GROUP BY c.state ORDER BY customer_count DESC
Output:
Showing first 10 rows out of 37 rows

state	customer_count
California	17
Texas	10
Florida	9
New York	7
Virginia	5
Michigan	5
Illinois	5
District of Columbia	5
Pennsylvania	4
Ohio	4

Observations and Insights:

- A total of 994 customers have placed orders for vehicles with the New-wheels company.
- Among these, California stands out as the state with the highest customer count, contributing 17 customers. Following California, Texas and Florida rank second and third, with 10 and 9 customers, respectively.
- In contrast, several states have a significantly lower customer presence, each with only 1 customer. These include Arizona, Nebraska, Idaho, New Hampshire, New Jersey, North Dakota, Oklahoma, and West Virginia.
- This distribution identifies potential areas for growth in states with minimal customer engagement by implementing targeted marketing campaigns, community engagement and customizing vehicle offerings based on regional preferences.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
    p.vehicle_maker, COUNT(DISTINCT o.customer_id) AS customer_count
FROM product_t AS p
INNER JOIN order_t AS o
    ON p.product_id = o.product_id
GROUP BY p.vehicle_maker
ORDER BY customer_count DESC
LIMIT 5;
```

Output:

SQL query passed

Query 1 **Result**: SELECT p.vehicle_maker, COUNT(DISTINCT o.customer_id) AS customer_count FROM product_t AS p JOIN order_t AS o ON p.product_id = o.product_id GROUP BY p.vehicle_maker ORDER BY customer_count DESC LIMIT 5

Output:

Showing 5 rows

vehicle_maker	customer_count
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

- Chevrolet emerges as the most favoured vehicle manufacturer, with approximately 83 customers, indicating its strong appeal among customers. Following closely is Ford, which has 63 customers, reinforcing its position as a popular choice.
- In the next tier, Toyota, Pontiac, and Dodge are nearly tied in sales, with 52, 50, and 50 customers, respectively. This close proximity in numbers suggests that these brands are equally well-received by customers, each enjoying a solid level of popularity.
- Overall, the demand for these five vehicle makers: Chevrolet, Ford, Toyota, Pontiac and Dodge dominates the market, highlighting a clear preference among consumers for these brands.
- Offering a wider range of models and customization options helps to cater diverse customer preferences for increasing the sales of least preferred cars.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT state, vehicle_maker, customer_count, rnk
FROM (
    SELECT
        c.state,
        p.vehicle_maker,
        COUNT(*) AS customer_count,
        RANK() OVER (PARTITION BY c.state ORDER BY COUNT(*) DESC) AS rnk
    FROM customer_t AS c
    INNER JOIN order_t AS o
        ON c.customer_id = o.customer_id
    INNER JOIN product_t AS p
        ON o.product_id = p.product_id
    GROUP BY c.state, p.vehicle_maker
) AS ranked_makers
WHERE rnk = 1
```

Output:

SQL query passed

Query 1 Passed: SELECT state, vehicle_maker, customer_count, rnk FROM (SELECT c.state, p.vehicle_maker, COUNT(*) AS customer_count, RANK() OVER (PARTITION BY c.state ORDER BY COUNT(*) DESC) AS rnk FROM customer_t AS c INNER JOIN order_t AS o ON c.customer_id = o.customer_id INNER JOIN product_t AS p ON o.product_id = p.product_id GROUP BY c.state, p.vehicle_maker) AS ranked_makers WHERE rnk = 1 ORDER BY state

Output:

Showing first 10 rows out of 101 rows

state	vehicle_maker	customer_count	rnk
Alabama	Lincoln	1	1
Alabama	Lexus	1	1
Alabama	Chevrolet	1	1
Arizona	Chevrolet	1	1
Arkansas	Pontiac	1	1
Arkansas	GMC	1	1
California	Pontiac	2	1
California	Nissan	2	1
California	Ford	2	1
California	Chevrolet	2	1

SQL queries passed

Query 1 **Passed**: SELECT state, vehicle_maker, customer_count, rnk FROM state
 Output:
 Showing first 10 rows out of 101 rows

state	vehicle_maker	customer_count	rnk
Alabama	Lincoln	1	1
Alabama	Lexus	1	1
Alabama	Chevrolet	1	1
Arizona	Chevrolet	1	1
Arkansas	Pontiac	1	1
Arkansas	GMC	1	1
California	Pontiac	2	1
California	Nissan	2	1
California	Ford	2	1
California	Chevrolet	2	1

Observations and Insights:

- The data indicates a varied preference for vehicle makers across states, suggesting that consumer choices are influenced by regional factors.
- In Alabama, customers favor Lincoln, Lexus, and Chevrolet. In Arizona, Chevrolet is the preferred choice. California customers show a preference for Pontiac, Nissan, Ford, and Chevrolet. In Florida, Ford and Volvo are popular among customers, while in Texas, Nissan is the top choice.
- Each state has shown a unique preference for specific vehicle makers, with some brands capturing the loyalty of individual customers.
- States with only 1 or 2 customers for each vehicle maker may represent untapped markets. Focused marketing efforts could help increase brand visibility and attract new customers.
- Developing targeted campaigns and highlighting the benefits and features of the preferred vehicle makers in those regions help in attracting more customers.

Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Solution Query:

Overall average rating given by the customers:

```
SELECT AVG(numeric_rating) AS overall_average_rating
FROM (
    SELECT
        CASE customer_feedback
            WHEN 'Very Bad' THEN 1
            WHEN 'Bad' THEN 2
            WHEN 'Okay' THEN 3
            WHEN 'Good' THEN 4
            WHEN 'Very Good' THEN 5
            ELSE NULL
        END AS numeric_rating
    FROM order_t
) AS reviews;
```

Average rating in each quarter:

```
SELECT
    quarter_number, AVG(numeric_rating) AS average_rating
FROM (
    SELECT
        quarter_number,
        CASE customer_feedback
            WHEN 'Very Bad' THEN 1
            WHEN 'Bad' THEN 2
            WHEN 'Okay' THEN 3
            WHEN 'Good' THEN 4
            WHEN 'Very Good' THEN 5
            ELSE NULL
        END AS numeric_rating
    FROM order_t
) AS reviews
GROUP BY quarter_number
ORDER BY quarter_number;
```


Output:

Overall average rating given by the customers:

SQL query passed

Query 1 **Passed**: SELECT AVG(numeric_rating) AS overall_average_rating FROM (SELECT CASE customer_feedback WHEN 'Very Bad' THEN 1 WHEN 'Bad' THEN 2 WHEN 'Okay' THEN 3 WHEN 'Good' THEN 4 WHEN 'Very Good' THEN 5 ELSE NULL END AS numeric_rating FROM order_t) AS rated_reviews

Output:
Showing 1 rows

overall_average_rating
3.135

Average rating in each quarter:

SQL query passed

Query 1 **Passed**: SELECT quarter_number, AVG(numeric_rating) AS average_rating FROM (SELECT quarter_number, CASE customer_feedback WHEN 'Very Bad' THEN 1 WHEN 'Bad' THEN 2 WHEN 'Okay' THEN 3 WHEN 'Good' THEN 4 WHEN 'Very Good' THEN 5 ELSE NULL END AS numeric_rating FROM order_t) AS rated_reviews GROUP BY quarter_number ORDER BY quarter_number

Output:
Showing 4 rows

quarter_number	average_rating
1	3.554838709677419
2	3.354961832061069
3	2.956331877292578
4	2.3969849246231156

Observations and Insights:

- The overall average customer rating for New-wheels company is 3.135, indicating a satisfactory level of service.
- However, the average rating declines over the quarters, starting at 3.55 in the first quarter and dropping to 2.39 by the fourth quarter, reflecting growing customer dissatisfaction over time.
- To enhance the customer experience and achieve a rating of 4, it is essential to improve management practices and the quality of service provided by the company.
- It is necessary to identify the areas for improvement and innovation in the management by gathering comprehensive customer feedback.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
    quarter_number,
    COUNT(CASE WHEN customer_feedback = 'Very Bad' THEN 1 END) * 100.0 / COUNT(*)
AS Percentage_Very_bad,
    COUNT(CASE WHEN customer_feedback = 'Bad' THEN 1 END) * 100.0 / COUNT(*) AS
Percentage_Bad,
    COUNT(CASE WHEN customer_feedback = 'Okay' THEN 1 END) * 100.0 / COUNT(*) AS
Percentage_Okay,
    COUNT(CASE WHEN customer_feedback = 'Good' THEN 1 END) * 100.0 / COUNT(*) AS
Percentage_Good,
    COUNT(CASE WHEN customer_feedback = 'Very Good' THEN 1 END) * 100.0 / COUNT(*)
AS Percentage_Very_good
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

SQL query result

Query 1 **Success**: SELECT quarter_number, COUNT(CASE WHEN customer_feedback = 'Very Bad' THEN 1 END) * 100.0 / COUNT(*) AS Percentage_Very_bad, COUNT(CASE WHEN customer_feedback = 'Bad' THEN 1 END) * 100.0 / COUNT(*) AS Percentage_Bad, COUNT(CASE WHEN customer_feedback = 'Okay' THEN 1 END) * 100.0 / COUNT(*) AS Percentage_Okay, COUNT(CASE WHEN customer_feedback = 'Good' THEN 1 END) * 100.0 / COUNT(*) AS Percentage_Good, COUNT(CASE WHEN customer_feedback = 'Very Good' THEN 1 END) * 100.0 / COUNT(*) AS Percentage_Very_good FROM order_t GROUP BY quarter_number ORDER BY quarter_number

Output:

Showing 4 rows

quarter_number	Percentage_Very_bad	Percentage_Bad	Percentage_Okay	Percentage_Good	Percentage_Very_good
1	10.96774193548387	11.290322580645162	19.032258064516128	28.70967741935484	30
2	14.885496183206106	14.122137404580153	20.229007633587788	22.137404580152673	28.625954198473284
3	17.903930131004365	22.707423580786028	21.83406113537118	20.96069868995633	16.593886462882097
4	30.65326633165829	29.14572864321608	20.100502512562816	10.050251256281408	10.050251256281408

Observations and Insights:

- The data shows a significant shift in customer satisfaction ratings throughout the year. Initially, in the first quarter, approximately 30% of customers rated their experience as 'Good' or 'Very Good,' reflecting a relatively positive perception of the company's services.
- However, as the year progressed, this figure plummeted to just 10% by the fourth quarter. The ratings for 'Bad' and 'Very Bad' experiences saw a corresponding increase, rising from 10% in the first quarter to 30% by the fourth quarter.
- This shift indicates that a growing number of customers are experiencing dissatisfaction, which could stem from various factors such as declining service quality, unresolved complaints, or unmet promises.
- It underscores the urgent need for the company to identify the root causes of this decline and implement strategic improvements to enhance the customer experience. Addressing these issues proactively will be critical in reversing this trend and restoring customer confidence.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
    COUNT (DISTINCT order_id) AS Total_Order, quarter_number
FROM order_t
GROUP BY quarter_number;
```

Output:

SQL queries passed

Query 1 (Passed): SELECT COUNT (DISTINCT order_id) AS Total_Order, quarter_number FROM order_t GROUP BY quarter_number

Output:
Showing 4 rows

Total_Order	quarter_number
310	1
262	2
229	3
199	4

Observations and Insights:

- The New-wheels company experiences a significant boost in sales during the first quarter, with a total of 310 orders, accounting for 31% of all orders placed throughout the year. This strong performance sets a positive tone for the year.
- The second quarter follows with 262 orders, while the third quarter sees a slight decline to 229 orders.
- The fourth quarter records the lowest sales volume, with only 199 orders.
- This trend indicates a pronounced pattern where order volume is highest at the beginning of the year, gradually declining as the year progresses.
- Such a trend may suggest seasonal influences on customer purchasing behavior, emphasizing the importance of targeted marketing strategies at the start of the year to capitalize on this peak demand.

Question 7: Calculate the net revenue generated by the company.

What is the quarter-over-quarter % change in net revenue?

Solution Query:

Net revenue generated by the company:

```
SELECT SUM (quantity * (vehicle_price - (vehicle_price * discount))) AS net_revenue
FROM order_t;
```

Quarter-over-quarter % change in net revenue:

```
SELECT
    quarter_number,
    SUM(quantity * (vehicle_price - (vehicle_price * discount))) AS net_revenue,
    LAG(SUM(quantity * (vehicle_price - (vehicle_price * discount)))) OVER (ORDER
BY quarter_number) AS previous_revenue,
    CASE
        WHEN LAG(SUM(quantity * (vehicle_price - (vehicle_price * discount)))) OVER
(OORDER BY quarter_number) IS NOT NULL THEN
            ((SUM(quantity * (vehicle_price - (vehicle_price * discount))) -
                LAG(SUM(quantity * (vehicle_price - (vehicle_price * discount))))
OVER (ORDER BY quarter_number)) /
                LAG(SUM(quantity * (vehicle_price - (vehicle_price * discount)))) OVER
(OORDER BY quarter_number)) * 100
            ELSE NULL
        END AS percent_change
FROM
    order_t
GROUP BY
    quarter_number
ORDER BY
    quarter_number;
```

Output:

Net revenue generated by the company:

SQL queries passed

Query 1 **Passed**: SELECT SUM (quantity * (vehicle_price - (vehicle_price * discount))) AS net_revenue FROM order_t
Output:
Showing 1 rows

net_revenue
48610993.7813

Quarter-over-quarter % change in net revenue:

SQL query used

Query 1 **Answer**: SELECT quarter_number, SUM(quantity * (vehicle_price - (vehicle_price * discount))) AS net_revenue, LAG(SUM(quantity * (vehicle_price - (vehicle_price * discount)))) OVER (ORDER BY quarter_number) AS previous_revenue, CASE WHEN LAG(SUM(quantity * (vehicle_price - (vehicle_price * discount)))) OVER (ORDER BY quarter_number) IS NOT NULL THEN ((SUM(quantity * (vehicle_price - (vehicle_price * discount))) - LAG(SUM(quantity * (vehicle_price - (vehicle_price * discount)))) OVER (ORDER BY quarter_number)) / LAG(SUM(quantity * (vehicle_price - (vehicle_price * discount)))) OVER (ORDER BY quarter_number)) * 100 ELSE NULL END AS percent_change FROM order_t GROUP BY quarter_number ORDER BY quarter_number

Output:

Showing 4 rows

quarter_number	net_revenue	previous_revenue	percent_change
1	18032549.899600014		
2	13122995.756199999	18032549.899600014	-27.22606714377602
3	8882298.844900003	13122995.756199999	-32.31500634522774
4	8573149.280599996	8882298.844900003	-3.48051298091048

Observations and Insights:

- The company generated a total revenue of \$48,610,993.78.
- An analysis of quarterly revenue reveals a peak in the first quarter, followed by a decline in subsequent quarters.
- Compared to previous revenues, the current revenue shows a decrease of 27% in the second quarter, 32% in the third quarter, and 3.5% in the fourth quarter.
- This consistent downward trend in both previous and current revenue highlights the need for strategic measures to boost sales.
- Additionally, addressing the drop in current revenue is equally important to ensure that the company not only recovers but also grows its net revenue sustainably in the future.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
    quarter_number, SUM (quantity * (vehicle_price - (vehicle_price * discount)))
AS net_revenue, COUNT (DISTINCT order_id) AS Total_Order
FROM order_t
GROUP BY quarter_number;
```

Output:

SQL queries passed

Query 1 (Passed): SELECT quarter_number, SUM (quantity * (vehicle_price - (vehicle_price * discount))) AS net_revenue, COUNT (DISTINCT order_id) AS Total_Order FROM order_t GROUP BY quarter_number

Output:

Showing 4 rows

quarter_number	net_revenue	Total_Order
1	18032549.899600014	310
2	13122995.756199999	262
3	8882298.844900003	229
4	8573149.280599996	199

Observations and Insights:

- Net revenue for the New-wheels company peaks in the first quarter, reaching \$18,032,549, driven by the highest order volume of 310 vehicles.
- The second quarter follows with a notable revenue of \$13,122,995.
- As we move into the third and fourth quarters, net revenue declines significantly, with the third quarter generating \$8,882,298 from 229 orders and the fourth quarter bringing in \$8,573,149 from 199 orders.
- This trend illustrates a consistent decrease in both net revenue and order volume across the quarters.
- The revenues from the third and fourth quarters are nearly comparable, indicating a stagnation in sales during this period.
- This pattern suggests the need for stronger marketing campaigns and strategic initiatives to increase vehicle sales in the latter half of the year, with the goal of maximizing any remaining demand.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT c.credit_card_type, AVG(o.discount) AS AVERAGE_DISCOUNT
FROM customer_t AS c
INNER JOIN order_t AS o
    ON c.customer_id = o.customer_id
GROUP BY c.credit_card_type
ORDER BY AVERAGE_DISCOUNT DESC;
```

Output:

SQL query passed

Query 1 **Success**: SELECT c.credit_card_type, AVG(o.discount) AS AVERAGE_DISCOUNT FROM customer_t AS c INNER JOIN order_t AS o ON c.customer_id = o.customer_id GROUP BY c.credit_card_type ORDER BY AVERAGE_DISCOUNT DESC
Output:
Showing first 10 rows out of 16 rows

credit_card_type	AVERAGE_DISCOUNT
instapayment	0.77
solo	0.7
americanexpress	0.6825000000000001
diners-club-enroute	0.665
diners-club-carte-blanc	0.6475000000000001
mastercard	0.64625
visa-electron	0.6380000000000001
maestro	0.6371428571428571
china-unionpay	0.624
laser	0.6219999999999999

Observations and Insights:

- Among the 16 credit cards analyzed, the instapayment and solo credit cards stand out with the highest average discounts, offering approximately 0.77 and 0.70, respectively.
- Following closely behind is the americanexpress card, providing an average discount of 0.682, and the Diners Club Enroute card, which offers an average discount of 0.665.
- On the lower end of the spectrum, the diners club us ca card provides a discount of 0.5, while the visa card offers an average discount of 0.565, marking them as the least favourable options in terms of discounts.
- This data highlights the varying discount benefits associated with different credit cards, allowing customers to make informed choices based on their preferences for savings.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT AVG (JULIANDAY(ship_date) - JULIANDAY(order_date)) AS Average_time,  
quarter_number  
FROM order_t  
GROUP BY quarter_number;
```

Output:

SQL queries passed

Query 1 (Passed): SELECT AVG (JULIANDAY(ship_date) - JULIANDAY(order_date)) AS Average_time, quarter_number FROM order_t GROUP BY quarter_number

Output:
Showing 4 rows

Average_time	quarter_number
57.16774193548387	1
71.11068702290076	2
117.75545851528385	3
174.09547738693468	4

Observations and Insights:

- The average shipping time for placed orders shows a consistent increase as the year progresses.
- In the first quarter, orders are typically shipped within an average of 57 days. This duration rises to 71 days in the second quarter, 117 days in the third quarter, and escalates significantly to 174 days in the fourth quarter.
- This notable increase means that the average shipping time in the last quarter is nearly three times longer than in the first quarter, which is quite substantial.
- The extended shipping times during the fourth quarter may be attributed to several factors, including holiday-related disruptions and a decrease in vehicle orders, both of which can strain logistical operations.
- This trend underscores the need for New-wheels to assess their shipping processes and consider strategies to mitigate delays, especially in the latter part of the year when demand is lower and operational challenges may arise.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
48610993.7813	1000	994	3.135
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
8573149.280	199	97.964	21.5

Business Recommendations

- **Focus on Continuous Improvement:** To enhance business performance from the beginning to the end of the year, it's essential to implement added benefits that can drive vehicle sales, even during the last quarter.
- **Customer Interest Analysis:** Analyzing customer interests by state and understanding vehicle preferences is crucial. By launching vehicles with additional options that cater to these preferences, we can stimulate customer interest and boost sales across all states.
- **Importance of Customer Feedback:** Customer feedback must be taken seriously to improve net revenue. With overall ratings currently just satisfactory, gathering detailed feedback will help identify areas that require more attention and improvement.
- **Shipping and Delivery Concerns:** Addressing shipping issues is vital. Timely vehicle delivery plays a significant role in building customer trust. Therefore, enhancing support and clarity in logistics and shipping processes is necessary.
- **Payment Options and Incentives:** Payment methods are a critical aspect to consider. Offering discounts through credit card promotions can motivate customers to purchase more vehicles. This strategy should be prioritized to maximize sales opportunities.